

An Introduction to Scientific Writing, Data Visualization, and Academic Publishing

Roman Zug

Part I: Scientific Writing

General structure of a scientific paper

Title

Abstract

Introduction

Methods

Results

Discussion

References

Title

- ***Fewest possible words that accurately describe the content of the paper***
- Be specific! Use important keywords (an improperly titled paper may never reach its intended audience)
- Avoid abbreviations
- Try to catch the reader's interest

Influence of aspirin on human megakaryocyte prostaglandin synthesis

Inhibition of prostaglandin synthesis as a mechanism of action of aspirin-like drugs

Treatment of polycystic ovary syndrome

How to treat . . .

The effect of calcium antagonist felodipine on blood pressure, heart rate, working capacity, plasma renin activity, plasma angiotensin II, urinary catecholamines and aldosterone in patients with essential hypertension

Essential hypertension: The effect of . . .

Abstract

- ***Summary of the paper***
- Typically not exceeding 250 words
- Self-contained
- Write the abstract last

Abstract: an annotated example

Annotated example taken from *Nature* 435, 114–118 (5 May 2005).

General background	One or two sentences providing a basic introduction to the field, comprehensible to a scientist in any discipline.	During cell division, mitotic spindles are assembled by microtubule-based motor proteins^{1,2}. The bipolar organization of spindles is essential for proper segregation of chromosomes, and requires plus-end-directed homotetrameric motor proteins of the widely conserved kinesin-5 (BimC) family³. Hypotheses for bipolar spindle formation include the ‘push–pull mitotic muscle’ model, in which kinesin-5 and opposing motor proteins act between overlapping microtubules^{2,4,5}. However, the precise roles of kinesin-5 during this process are unknown. Here we show that the vertebrate kinesin-5 Eg5 drives the sliding of microtubules depending on their relative orientation. We found in controlled <i>in vitro</i> assays that Eg5 has the remarkable capability of simultaneously moving at $\sim 20 \text{ nm s}^{-1}$ towards the plus-ends of each of the two microtubules it crosslinks. For anti-parallel microtubules, this results in relative sliding at $\sim 40 \text{ nm s}^{-1}$, comparable to spindle pole separation rates <i>in vivo</i>⁶. Furthermore, we found that Eg5 can tether microtubule plus-ends, suggesting an additional microtubule-binding mode for Eg5. Our results demonstrate how members of the kinesin-5 family are likely to function in mitosis, pushing apart interpolar microtubules as well as recruiting microtubules into bundles that are subsequently polarized by relative sliding. We anticipate our assay to be a starting point for more sophisticated <i>in vitro</i> models of mitotic spindles. For example, the individual and combined action of multiple mitotic motors could be tested, including minus-end-directed motors opposing Eg5 motility. Furthermore, Eg5 inhibition is a major target of anti-cancer drug development, and a well-defined and quantitative assay for motor function will be relevant for such developments.
Specific background	Two to three sentences of more detailed background, comprehensible to scientists in related disciplines.	
Knowledge gap	One sentence clearly stating the general problem being addressed by this particular study.	
“Here we show...”	One sentence summarizing the main result (with the words “here we show” or their equivalent).	
Key results with concrete values/statements	Two or three sentences explaining what the main result reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge.	
Significance of results	One or two sentences to put the results into a more general context.	
	Two or three sentences to provide a broader perspective, readily comprehensible to a scientist in any discipline, may be included in the first paragraph if the editor considers that the accessibility of the paper is significantly enhanced by their inclusion. Under these circumstances, the length of the paragraph can be up to 300 words. (This example is 190 words without the final section, and 250 words with it).	

Introduction

- Present the problem and the proposed solution (Why was there a need to conduct the study?)
- Move from general to specific
- Clearly state scope and objectives (optional: brief statement of principle findings)
- Introduce the pertinent literature
 - Be specific!
 - Don't introduce the authors, but their work

“Parmenter (1976) and Chessman (1978) studied the diet of *Chelodina longicollis* at various latitudes, and Legler (1978) and Chessman (1983) conducted a similar study on *Chelodina expansa*.”

“*Chelodina expansa* is a selective and specialised predator feeding upon motile prey such as decapod crustaceans, aquatic bugs and small fish (Legler, 1978; Chessman, 1983), whereas *C. longicollis* has a diverse and opportunistic diet (Parmenter, 1976; Chessman, 1978).”

Introduction

THE NEW ENGLAND JOURNAL OF MEDICINE

Dec. 11, 1975

MEDICAL OBFUSCATION: STRUCTURE AND FUNCTION

Most medical communications are difficult to read. To determine why, contributions to three issues of the *New England Journal of Medicine* were studied, and the prose analyzed.

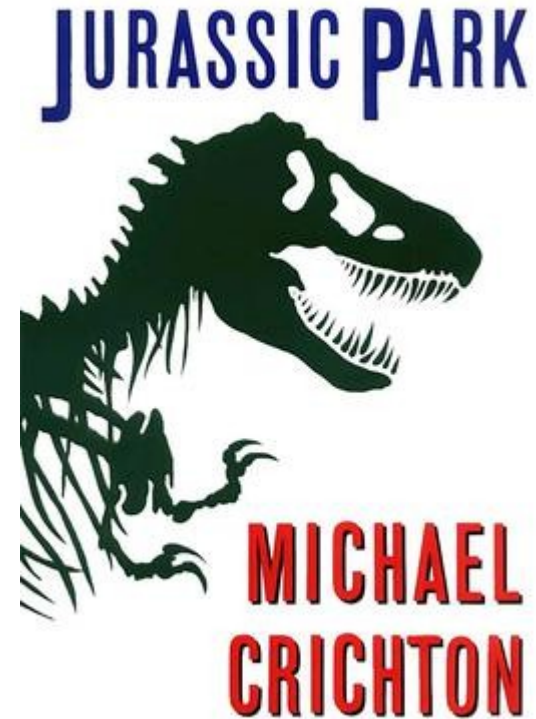
MATERIALS AND METHODS

Articles were taken from the *Journal* issues dated April 4, 1974; February 6, 1975; and October 16, 1975. These issues lay buried under a pile of papers on my desk. Articles were read at random.

RESULTS

I identified 10 recurring faults in the *Journal* articles I read.

MICHAEL CRICHTON, M.D.



Methods

- ***Main purpose: provide enough detail for a competent researcher to repeat your study and reproduce the results***
- Exact technical specifications of methods, materials, quantities, study site, etc.
- Chronological presentation
- Theoretical papers: include equations/math, but not the code
- Due to space constraints, part of the Methods section can also go in the supplementary material
- Previously published methods can be referenced:

The kidney volume was measured *with an ultrasound apparatus containing a built-in volume program*.³

Results

- ***Present your findings***
- Figures and tables are central!
- Be specific, use simple and clear language

Do not say:

"It is clearly evident from Fig. 1 that bird species richness increased with habitat complexity".

Say instead

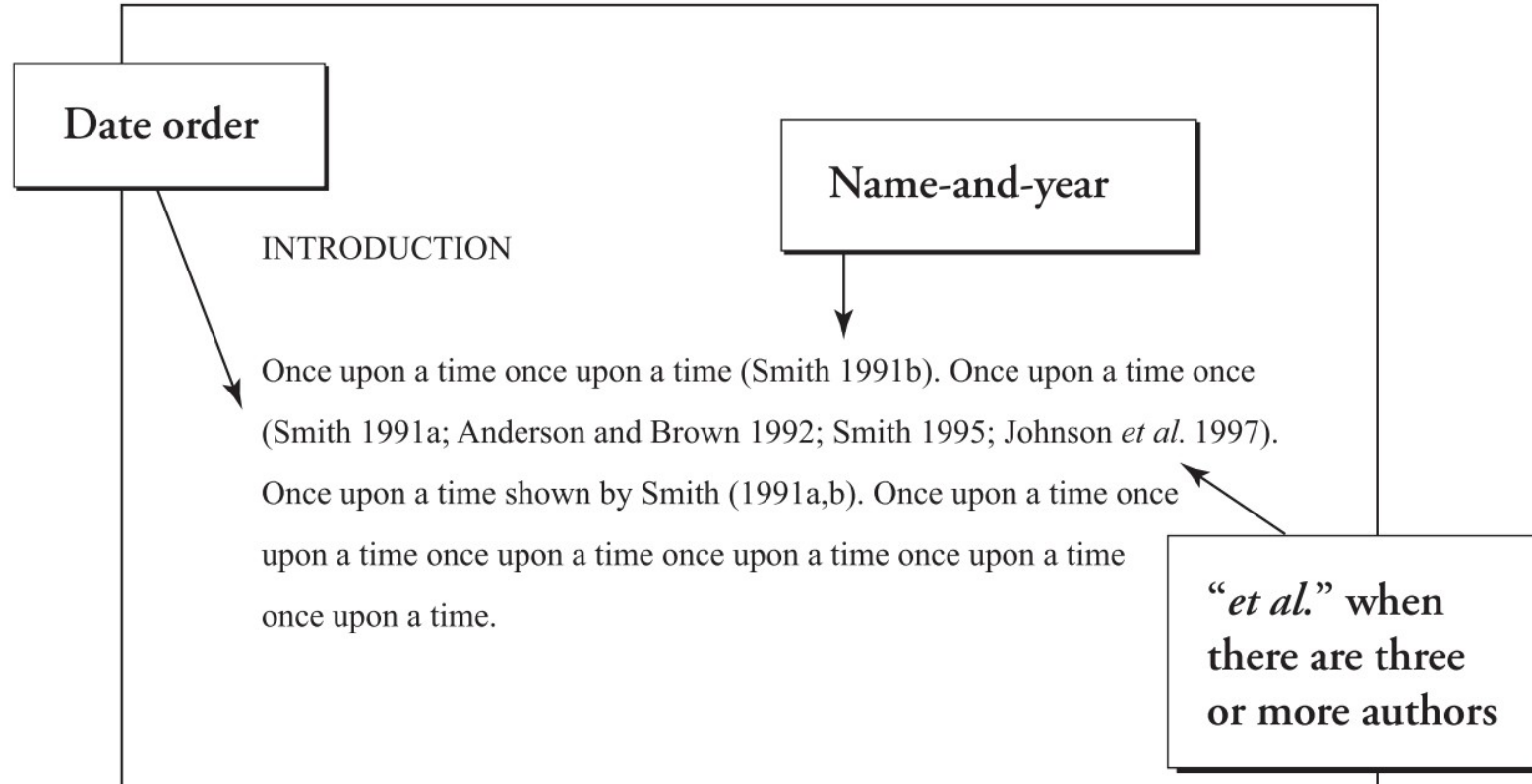
"Bird species richness increased with habitat complexity (Fig. 1)".

Discussion

- What do your results mean?
- How do the results compare to the findings of others or to expectations based on previous work? (cite references!)
- What conclusions can be drawn?
- How do the results fit into a broader context?
- Move from specific to general
- Also discuss problems, shortcomings, alternative explanations

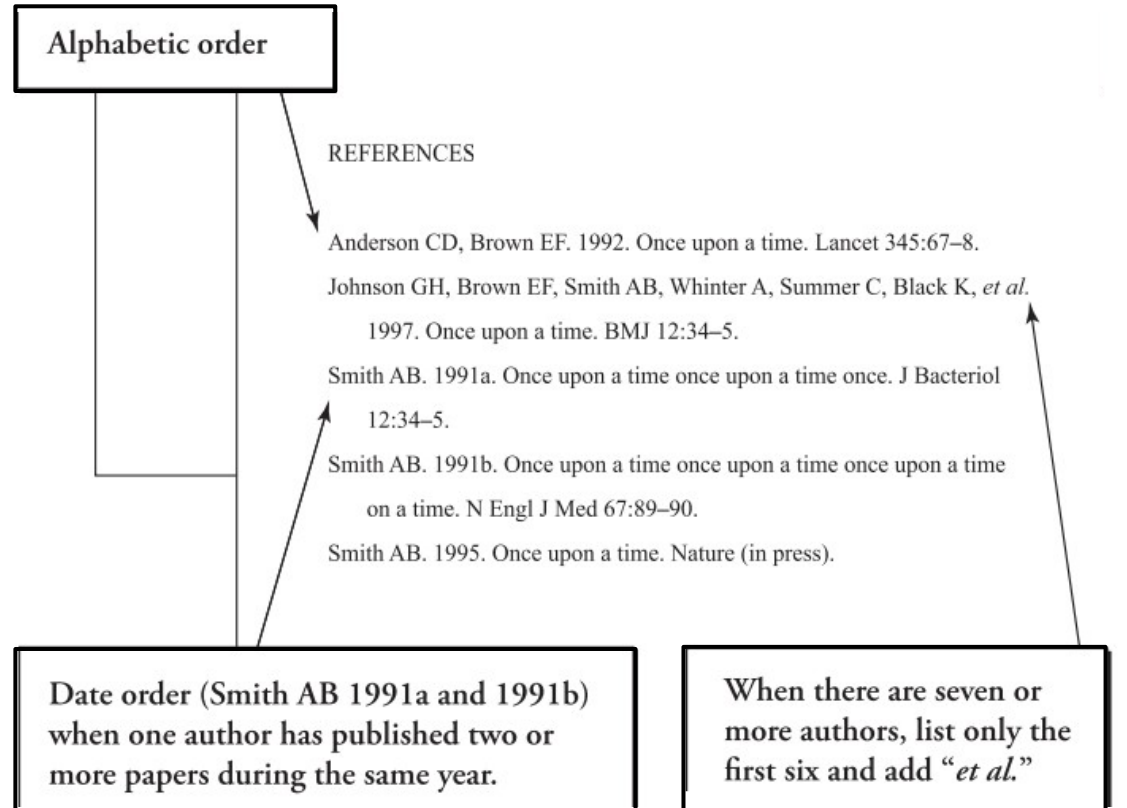
References

How are references cited in the main text?



References

How are references listed in the References section?



References

	Author	Title	Journal	Volume	
17.	Belikov, A. V.	Age-related diseases as vicious cycles.	<i>Ageing Res. Rev.</i>	49 ,	11–26.
	https://doi.org/10.1016/j.arr.2018.11.002	(2019).			
	DOI (digital object identifier)	Year			Page numbers

References

Two major styles of referencing

Author-year system

This method was introduced by Aburel in 1938, but he was followed by only a few workers in the succeeding 20 years (Bommelaer 1948; Cioc 1948; Kosowski 1949; de Watteville and d'Enst 1950). During the 1960's however, hypertonic saline has been increasingly employed (Bengtsson and Csapo 1962; Jaffin *et al.* 1962; Wagner *et al.* 1962; Larsson-Cohn 1964; Møller *et al.* 1964; Sciarra *et al.* 1964; Wiqvist and Eriksson 1964; Bora 1965; Short *et al.* 1965; Turnbull and Andersson 1965; Wagatsuma 1965; Cameron and Dayan 1966; Gochberg and Reid 1966; Klopper *et al.* 1966; Christie *et al.* 1966; Ruttner 1966; Olsen *et al.* 1967).

Numbering system

This method was introduced by Aburel in 1938,¹ but he was followed by only a few workers in the succeeding 20 years.²⁻⁵ During the 1960s, however, hypertonic saline has been increasingly employed.⁶⁻²²

The two major types of scientific articles

Research article

- Reports original research
- Based on the analysis and interpretation of this original data

Review article

- Does not report original research
- Based on existing literature, but may still provide a novel perspective

Perhaps the single most important writing advice

Omit useless words

Omit useless words

Do not say:

"It is clearly evident from Fig. 1 that bird species richness increased with habitat complexity".

Say instead

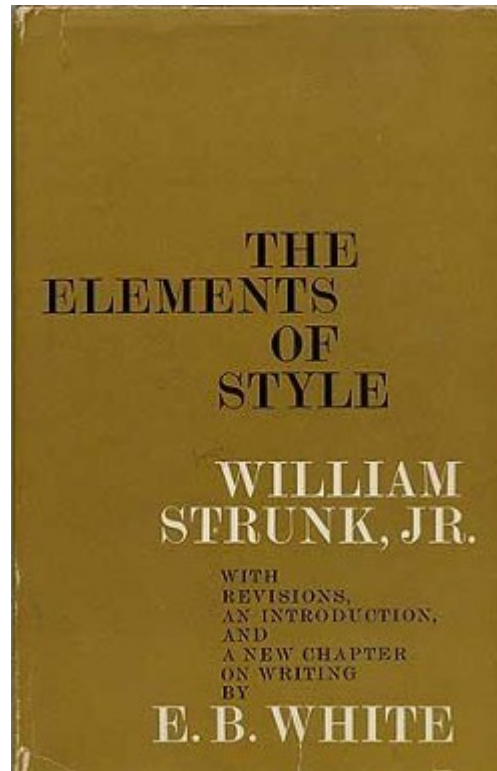
"Bird species richness increased with habitat complexity (Fig. 1)".

Omit useless words

Our research, designed to test the fatal effects of PGF2 α on dogs, was carried out by intravenously introducing the drug. In the experiments, a relatively small quantity, 30 mg, was administered to each animal. In each case, PGF2 α proved fatal; all 10 dogs expiring before a lapse of five minutes after the injection.

Intravenous injection of 30 mg prostaglandin PGF2 α to each of ten dogs killed them within five minutes.

Strunk & White, The Elements of Style (1959)



Active part:
Writing an abstract

Remember the lecture on sex ratio theory: *Wolbachia*-induced male killing

- *Wolbachia*: bacterial symbionts of insects; only transmitted via females
- Different strategies to induce female-biased sex ratios, incl. male killing
- *Wolbachia* kill male eggs so that sisters of killed males obtain a fitness benefit
- Results in extremely biased sex ratios (1 male per 100 females), e.g. in the butterfly *Hypolimnys bolina*
- Strong selection on butterflies to suppress male-killing
- Indeed, evolution of suppressor genes was found in the 2000s, "one of the fastest evolutionary changes ever observed"



Extraordinary Flux in Sex Ratio (Charlat et al., 2007)

BREVIA

Extraordinary Flux in Sex Ratio

Sylvain Charlat,^{1,2*} Emily A. Hornett,¹ James H. Fullard,³ Neil Davies,² George K. Roderick,⁴ Nina Wedell,⁵ Gregory D. D. Hurst¹

The ratio of males to females within a species is central to many aspects of its biology, from mating frequency (1) to population growth and disease spread. Forty years ago, Hamilton (2) noted that the evolutionary arms race engaged between "selfish" sex-ratio distorters and counteracting suppressor genes has the potential to produce rapid flux in sex ratio. Here we document this process in action in a tropical butterfly and report a transition from 99% females to parity within 10 generations.

The butterfly *Hypolimnarus bolina* is infected by a maternally inherited *Wolbachia* bacterium that selectively kills male embryos. Extreme population sex ratio bias associated with high prevalence of the male killer has been recorded in Polynesian populations, most notably on the Samoan islands Upolu and Savaii, where males represented only 1% of individuals in 2001 (3). In May and June 2005, informal observations revealed substantial numbers of males at three different locations on Upolu, suggesting dramatic alteration of the sex ratio. At the same time, the sex ratio on neighboring Savaii remained at >99% female, with no males observed (~100 individuals were observed on each island).

We carried out a formal survey of *H. bolina* sex ratio in 2006 and also scored male-killing efficiency

via the sex ratio produced by wild caught females (Fig. 1). On Upolu, where males were very rare in 2001 but common in 2005, we collected 20 males and 33 females in total in 2006 from two sites within the island. We reared progeny from 14 females, and all produced sons. This represents a marked difference from the 2001 data, where only 3 of 64 females gave rise to any male offspring ($\chi^2 = 55.76$, $df = 1$, $P < 0.0001$). Overall, the sex ratio produced by Upolu females was near 1:1 (83 females to 75 males), with no heterogeneity within or between sampling sites. Lack of male killing was also corroborated by high egg hatch rates.

The 2006 data from Savaii showed a similar transformation: Barely 10 generations after extreme female bias was reported, 21 males and 33 females were collected from Savaii. At Salelologa, close to Upolu, the shift to a 1:1 sex ratio was nearly complete: All females produced sons, there was no heterogeneity between broods, and the overall sex ratio produced was consistent with 1:1. High egg hatch rate also indicates loss of male killing. In contrast, at Sagone (more distant from Upolu), the shift to a 1:1 sex ratio was incomplete: Three of six females produced no sons, and sex ratio was heterogeneous between broods. Con-

sistent with retention of male killing, the median egg hatch rate was 71% (Fig. 1).

Polymerase chain reaction (PCR) assay of the bacterium revealed that all individuals from Upolu and Savaii, male and female, carry *Wolbachia*, and sequence identity at six *Wolbachia* loci confirmed that the previously present strain had not been replaced (4). Thus, the change in *H. bolina* sex ratios arose because of a loss of male-killing efficiency in the existing *Wolbachia* rather than either loss of infection or displacement by a new strain. We tested whether loss of efficiency was caused by the evolution of suppressor genes in the butterfly host, as previously observed (5). Introgressing the Samoan *Wolbachia* onto a nonsuppressor *H. bolina* nuclear background (4) restored full male-killing ability after three generations of introgression, demonstrating that the shift in sex ratio was caused by the spread of host suppressor genes (table S1 and Fig. S1).

These results establish that conflicts associated with cytoplasmic sex ratio distorters can generate very rapid shifts in local population sex ratio, in our case from 100:1 to near 1:1 in fewer than 10 generations. The nature of the processes targeted by sex ratio distorters (for example, sex determination) suggests that intense selection for suppression could drive the evolution of core developmental traits that otherwise remain stable over time. With *Wolbachia* being just one of many widespread sex ratio distorters, it is probable that such phenomena occurred in the past on many branches of the tree of life.

References and Notes

1. S. Charlat et al., *Cor. Biol.* 37, 273 (2007).
2. W. D. Hamilton, *Science* 156, 477 (1965).
3. E. A. Dyon, G. D. D. Hurst, *Proc. Natl. Acad. Sci. USA* 103, 6520 (2006).
4. Materials and methods are available on Science Online.
5. E. A. Hornett et al., *PLoS Biol.* 4, e283 (2006).
6. We thank C. Vermetot and A. Dugloy for assistance and O. Dutton and M. Bender for comments. Funded by NSF (USA, no. 0416248), Natural Environment Research Council (UK, no. N085032921), and Natural Sciences and Engineering Research Council (Canada). Sequences were deposited in GenBank with accession numbers EF599992 to EF599996.

Supporting Online Material

www.sciencemag.org/cgi/content/full/317/5835/14/DC1

Materials and Methods

Fig. S1

Table S1

References and Notes

3 April 2007; accepted 16 May 2007

10.1126/science.1143369

¹Department of Biology, University College London, 4 Stephenson Way, London NW1 2HE, UK. ²Group South Pacific Research Station, University of California, Berkeley, BP 244 Mahanui, 96728 Moorea, French Polynesia. ³Department of Biology, University of Toronto at Mississauga, 3359 Mississauga Road W., Mississauga, ON L5L 1C6, Canada. ⁴Environmental Science (ESPM), University of California, Berkeley, CA 94720-3114, USA. ⁵School of Biosciences, University of Exeter, Cornwall Campus, Tremough, Penryn TR20 2EZ, UK.

*To whom correspondence should be addressed. E-mail: s.charlat@ucl.ac.uk

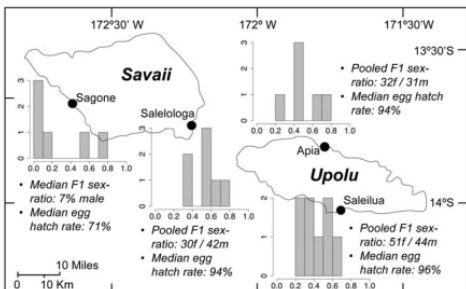


Fig. 1. F₁ sex ratio and egg hatch rate in 2006. Histograms give the distribution of F₁ sex ratios (x axis, proportion of males; y axis, number of broods). Sex ratios are homogeneous within sites at Apia ($\chi^2 = 7.59$, $df = 5$, $P = 0.18$), Saleilua ($\chi^2 = 5.68$, $df = 7$, $P = 0.58$), and Salelologa ($\chi^2 = 0.89$, $df = 6$, $P = 0.96$) and are compatible with a 1:1 sex ratio (Upolu pooled $\chi^2 = 0.41$, $df = 1$, $P = 0.52$; Salelologa $\chi^2 = 2$, $df = 1$, $P = 0.16$). Sex ratios are heterogeneous at Sagone ($\chi^2 = 18.27$, $df = 5$, $P < 0.003$). Breeding is as described in (2).

- One-page Science paper
- Write an abstract!

Annotated example taken from *Nature* 435, 114–118 (5 May 2005).

One or two sentences providing a basic introduction to the field, comprehensible to a scientist in any discipline.

Two to three sentences of more detailed background, comprehensible to scientists in related disciplines.

One sentence clearly stating the general problem being addressed by this particular study.

One sentence summarizing the main result (with the words "here we show" or their equivalent).

Two or three sentences explaining what the main result reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge.

One or two sentences to put the results into a more general context.

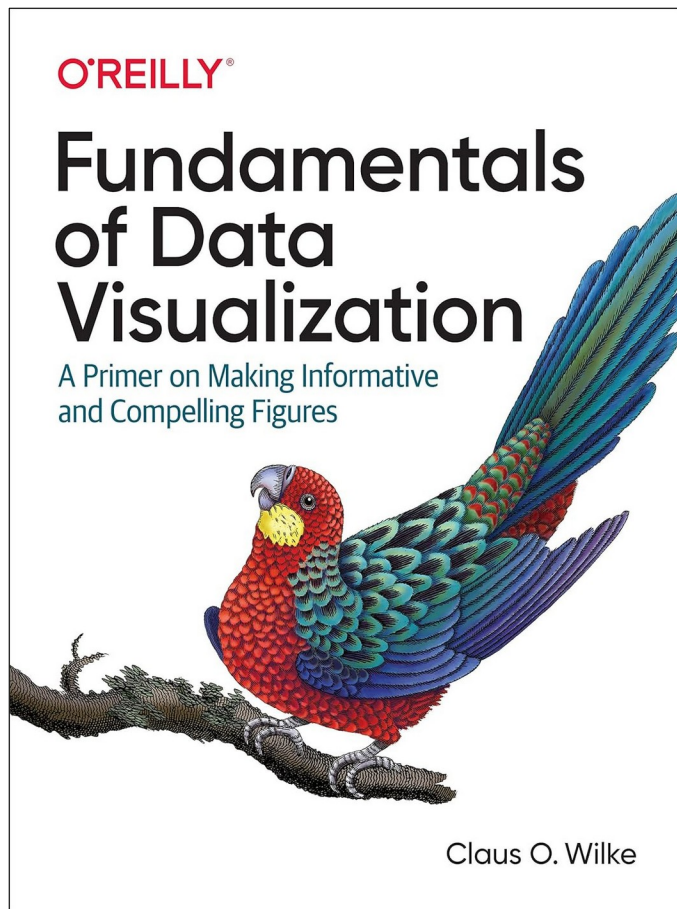
Two or three sentences to provide a broader perspective, readily comprehensible to a scientist in any discipline, may be included in the first paragraph if the editor considers that the accessibility of the paper is significantly enhanced by their inclusion. Under these circumstances, the length of the paragraph can be up to 300 words. (This example is 190 words without the final section, and 250 words with it).

During cell division, mitotic spindles are assembled by microtubule-based motor proteins^{1,2}. The bipolar organization of spindles is essential for proper segregation of chromosomes, and requires plus-end-directed homotetrameric motor proteins of the widely conserved kinesin-5 (BimC) family³. Hypotheses for bipolar spindle formation include the 'push-pull mitotic muscle' model, in which kinesin-5 and opposing motor proteins act between overlapping microtubules^{2,4,5}. However, the precise roles of kinesin-5 during this process are unknown. Here we show that the vertebrate kinesin-5 Eg5 drives the sliding of microtubules depending on their relative orientation. We found in controlled *in vitro* assays that Eg5 has the remarkable capability of simultaneously moving at ~20 nm s⁻¹ towards the plus-ends of each of the two microtubules it crosslinks. For anti-parallel microtubules, this results in relative sliding at ~40 nm s⁻¹, comparable to spindle pole separation rates *in vivo*⁶. Furthermore, we found that Eg5 can tether microtubule plus-ends, suggesting an additional microtubule-binding mode for Eg5. Our results demonstrate how members of the kinesin-5 family are likely to function in mitosis, pushing apart interpolar microtubules as well as recruiting microtubules into bundles that are subsequently polarized by relative sliding. We anticipate our assay to be a starting point for more sophisticated *in vitro* models of mitotic spindles. For example, the individual and combined action of multiple mitotic motors could be tested, including minus-end-directed motors opposing Eg5 motility. Furthermore, Eg5 inhibition is a major target of anti-cancer drug development, and a well-defined and quantitative assay for motor function will be relevant for such developments.

Part II:

Data Visualization

Claus O. Wilke, *Fundamentals of Data Visualization* (2019)



clauswilke.com/dataviz/

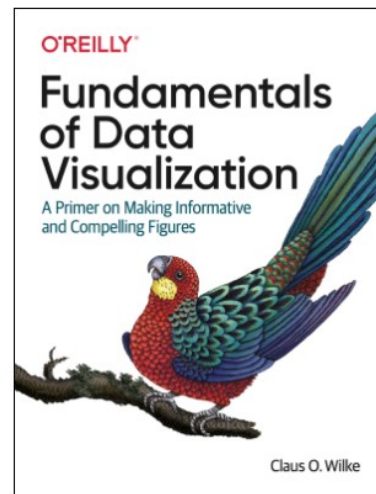
Fundamentals of Data Visualization

Claus O. Wilke

Welcome

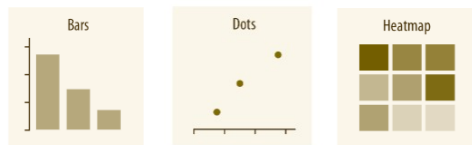
This is the website for the book "Fundamentals of Data Visualization," published by O'Reilly Media, Inc. The website contains the complete author manuscript before final copy-editing and other quality control. If you would like to order an official hardcopy or ebook, you can do so at various resellers, including [Amazon](#), [Barnes and Noble](#), [Google Play](#), or [Powells](#).

The book is meant as a guide to making visualizations that accurately reflect the data, tell a story, and look professional. It has grown out of my experience of working with students and postdocs in my laboratory on thousands of data visualizations. Over the years, I have noticed that the same issues arise over and over. I have attempted to collect my accumulated knowledge from these interactions in the form of this book.

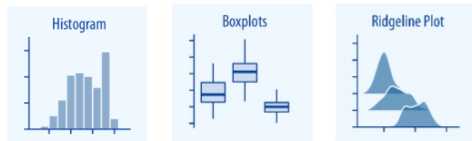


An overview of visualizations (non-exhaustive)

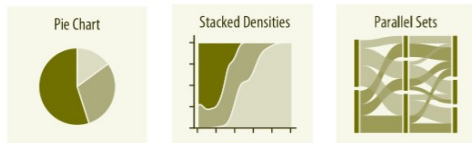
Amounts



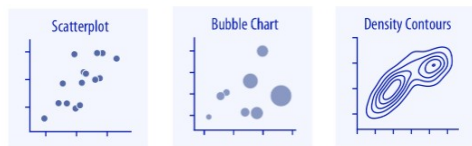
Distributions



Proportions



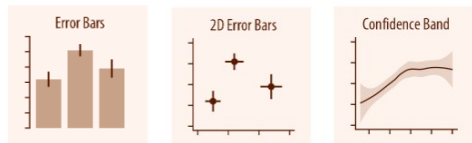
x-y relationships



Geospatial data

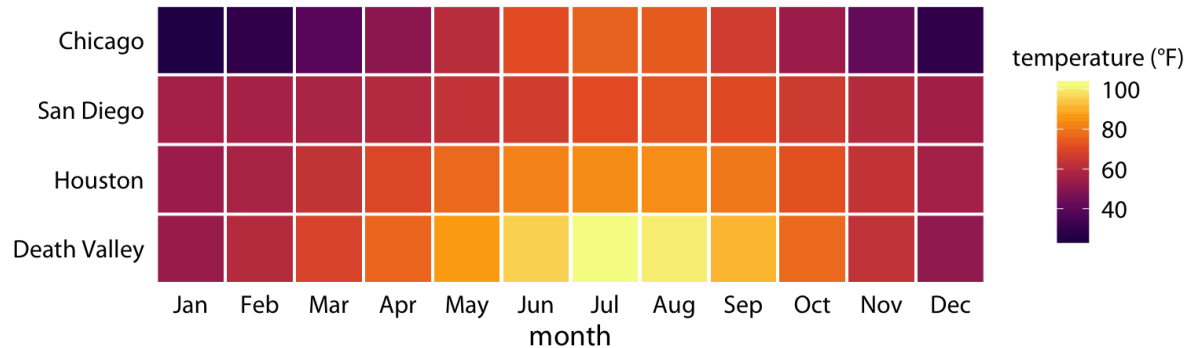
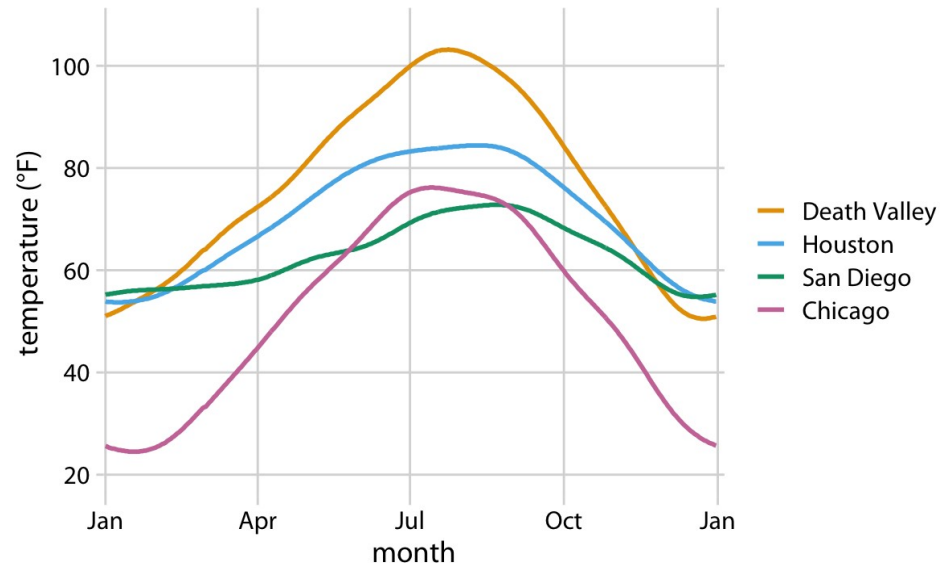


Uncertainty

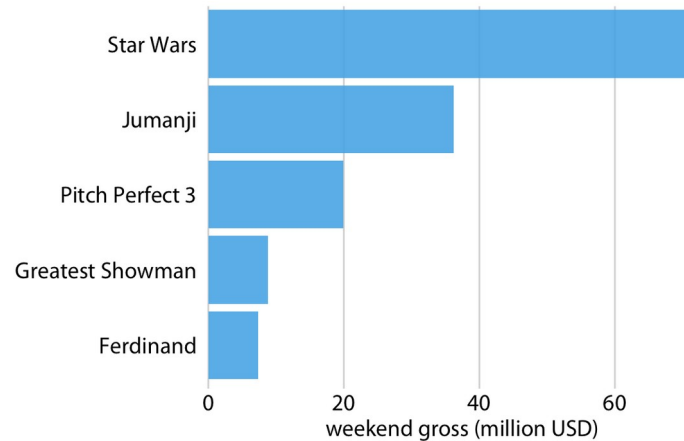
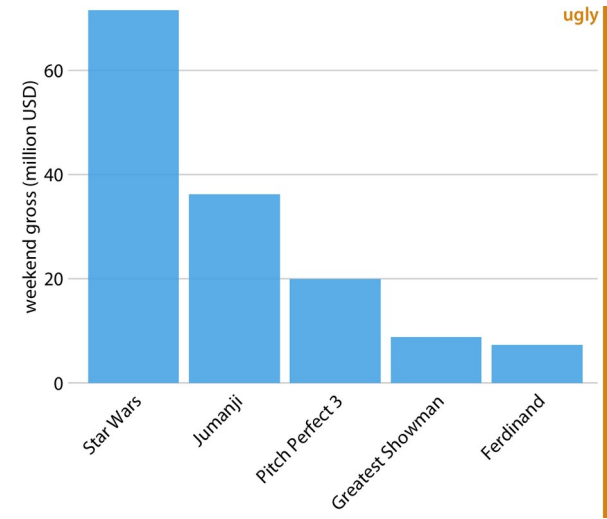
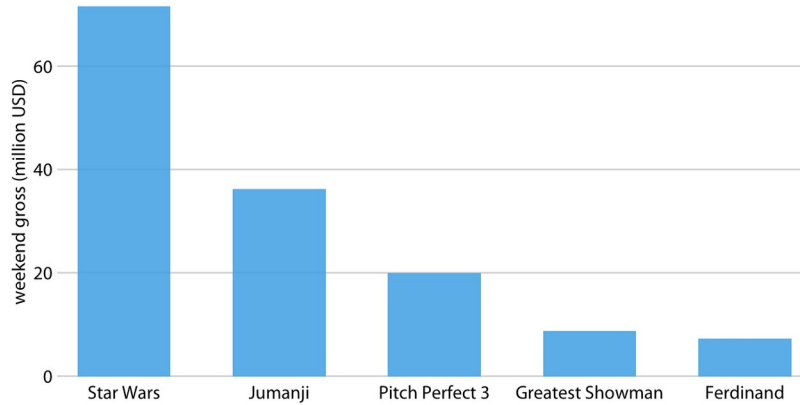


Pick the visualization that best fits your specific dataset and that highlights the key data features you want to show.

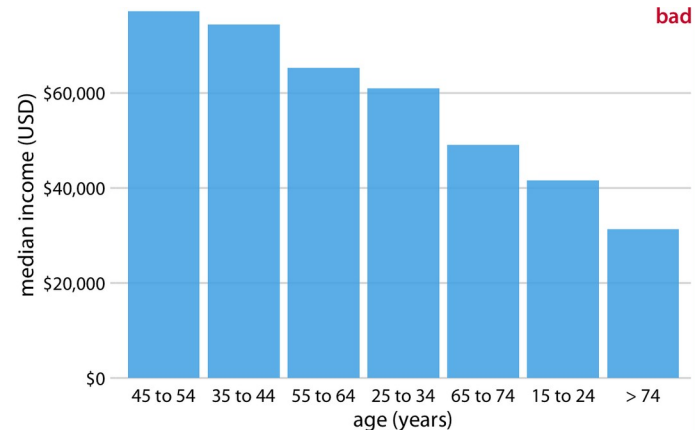
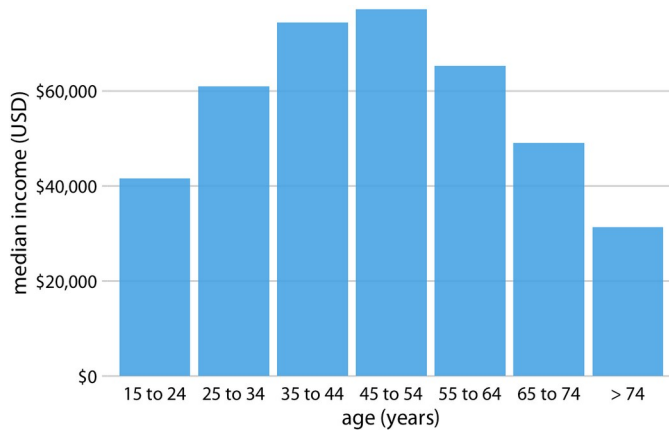
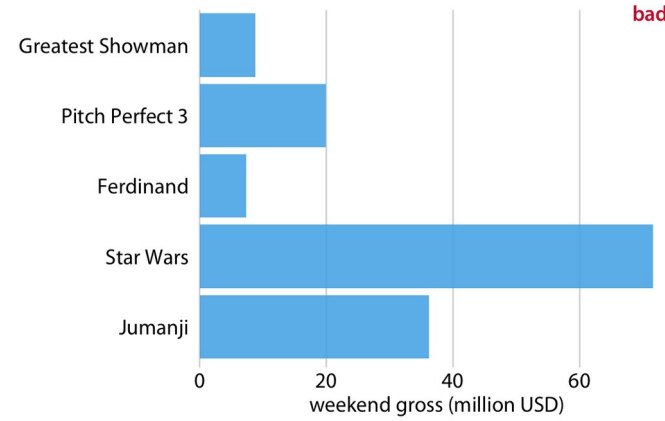
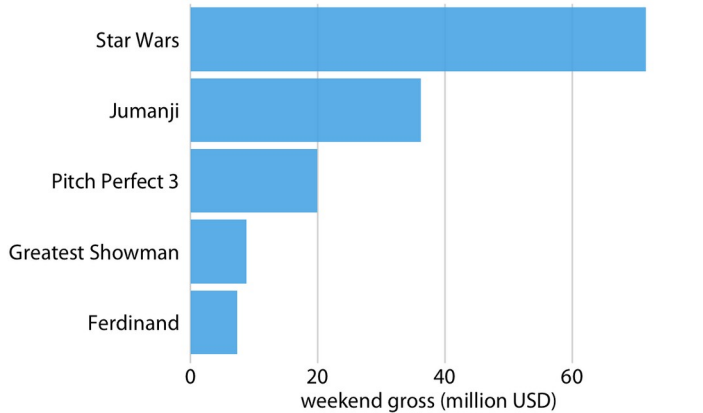
Use your freedom of visualization in a smart way!



Use your freedom of visualization in a smart way!



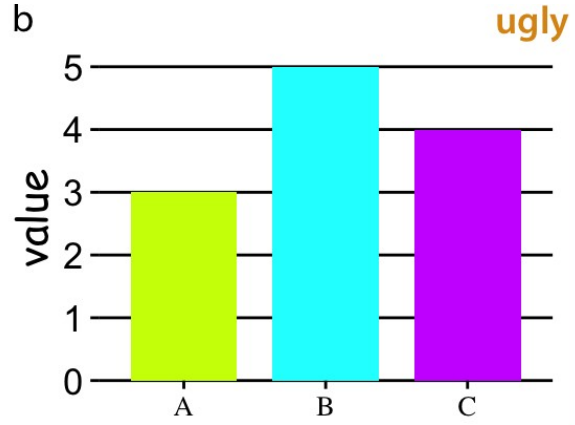
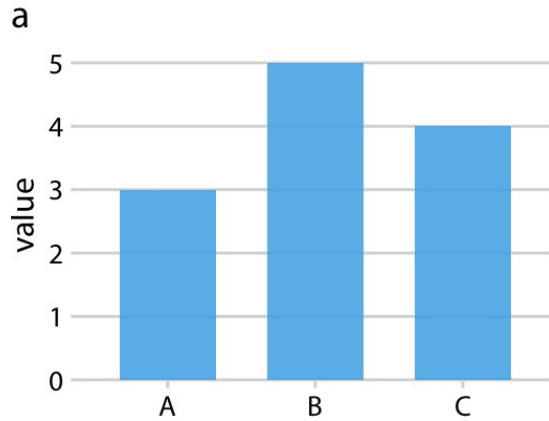
Pay attention to the order of the bars



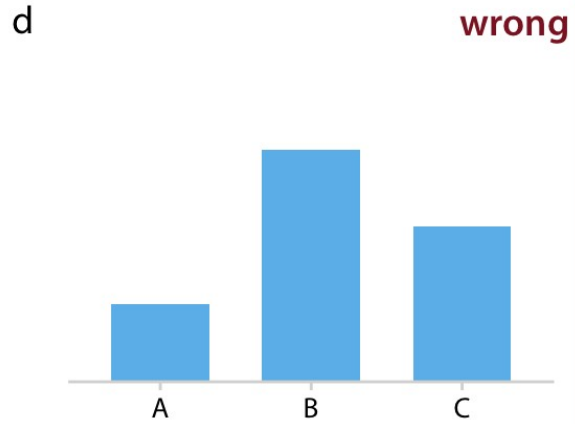
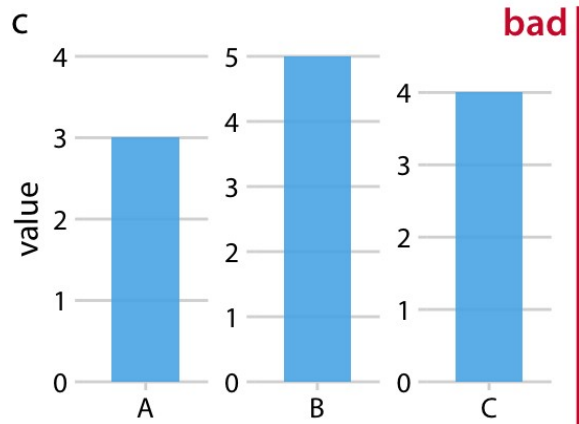
Wilke, 2019

If the bars represent unordered categories, order them by ascending or descending data values

Ugly, bad, and wrong figures



ugly: a figure that has aesthetic problems but otherwise is clear and informative.



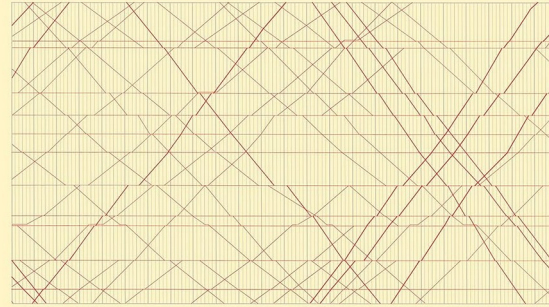
bad: a figure that has problems related to perception; it may be unclear, confusing, overly complicated, or deceiving.

wrong: a figure that has problems related to mathematics; it is objectively incorrect.

Perhaps the single most important guideline for visualizations

Create the simplest graph
that conveys the information
you want to convey

Edward Tufte, *The Visual Display of Quantitative Information* (1983)



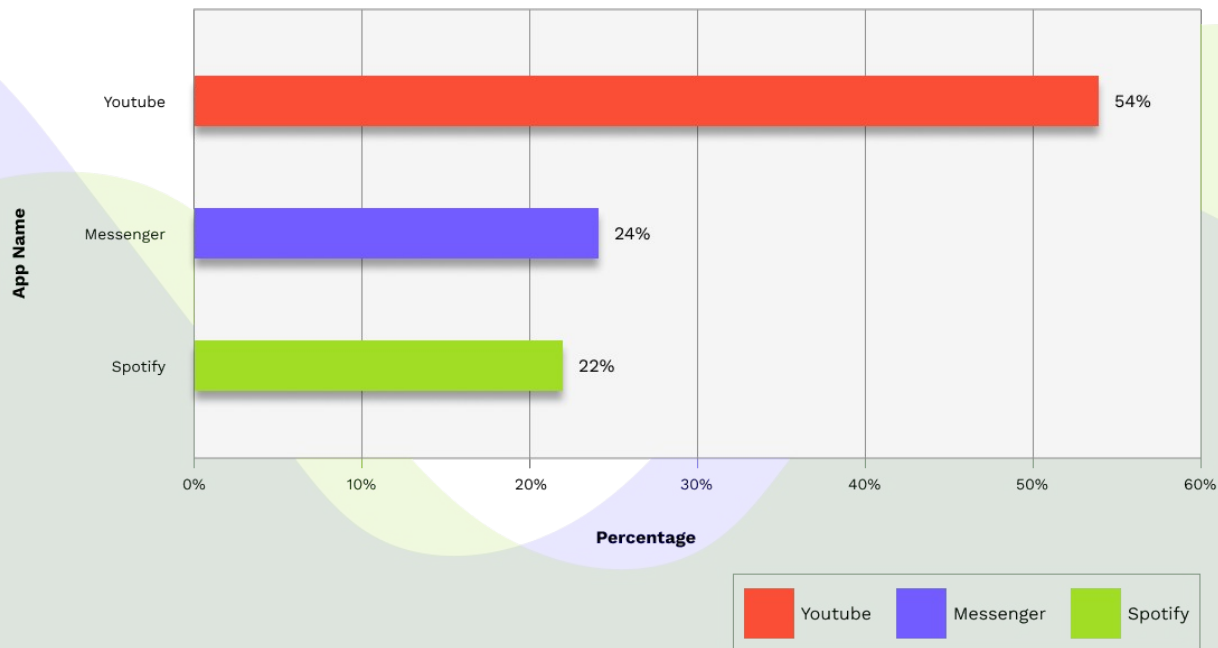
SECOND EDITION

The Visual Display of Quantitative Information

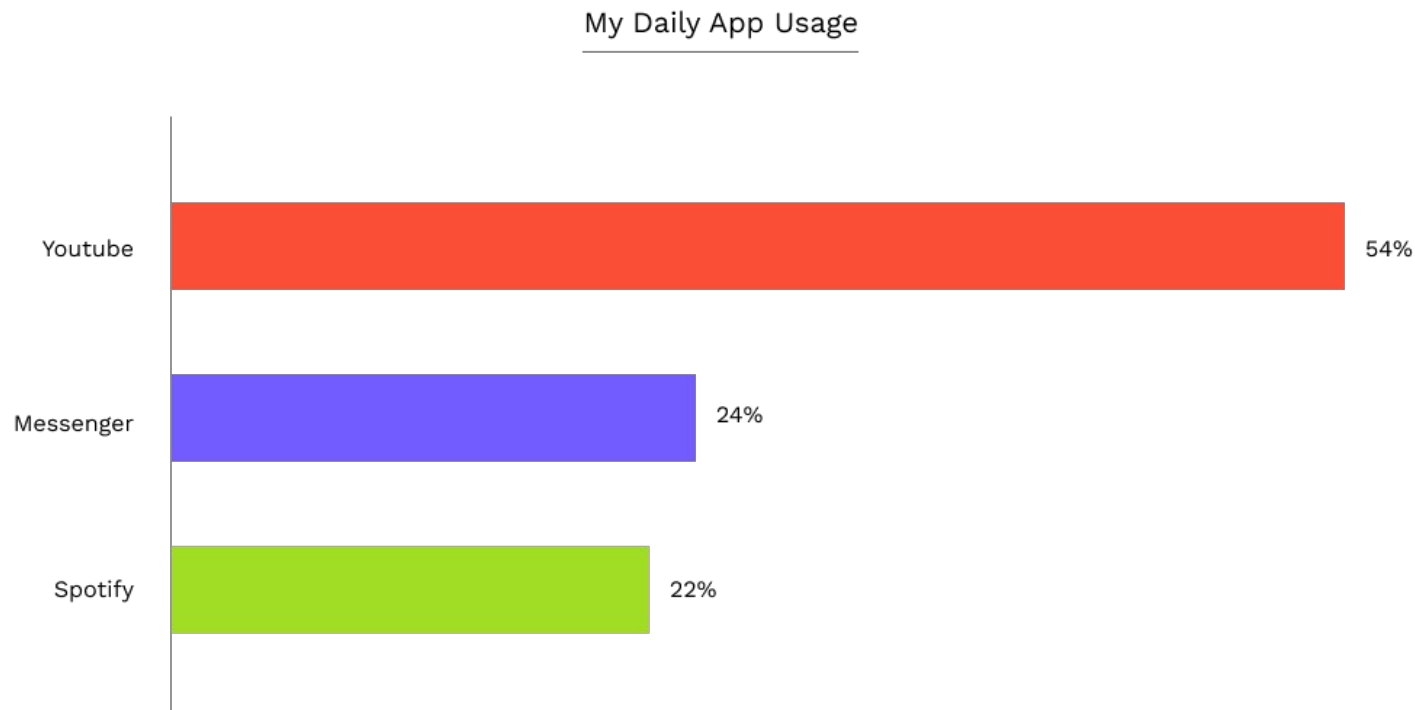
EDWARD R. TUFTE

Create the simplest graph that conveys the information you want to convey

My Daily App Usage

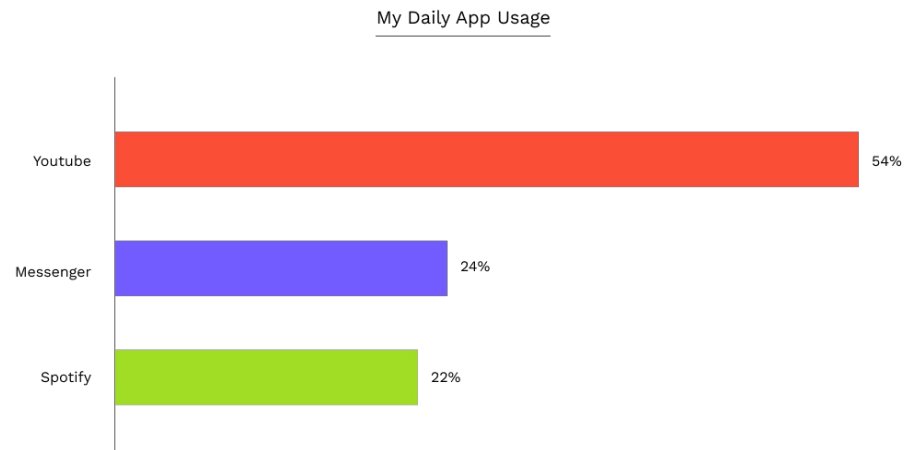
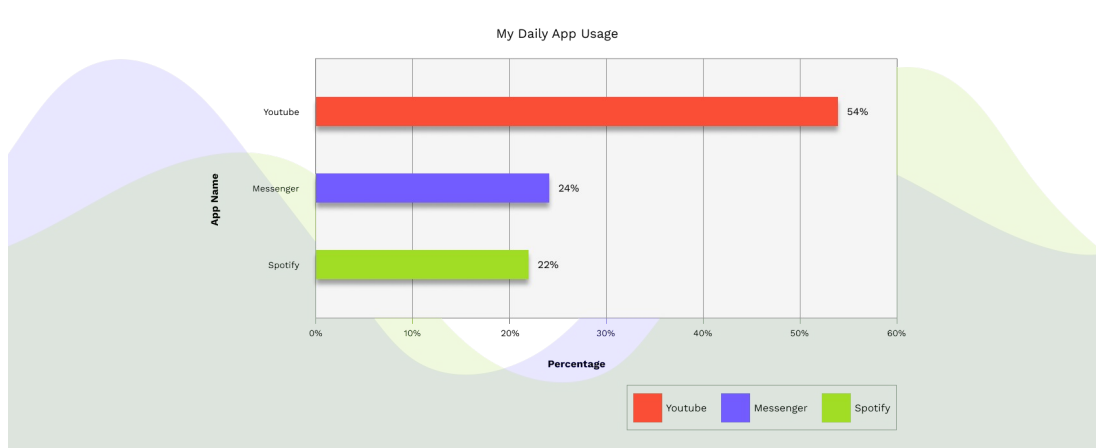


Create the simplest graph that conveys the information you want to convey



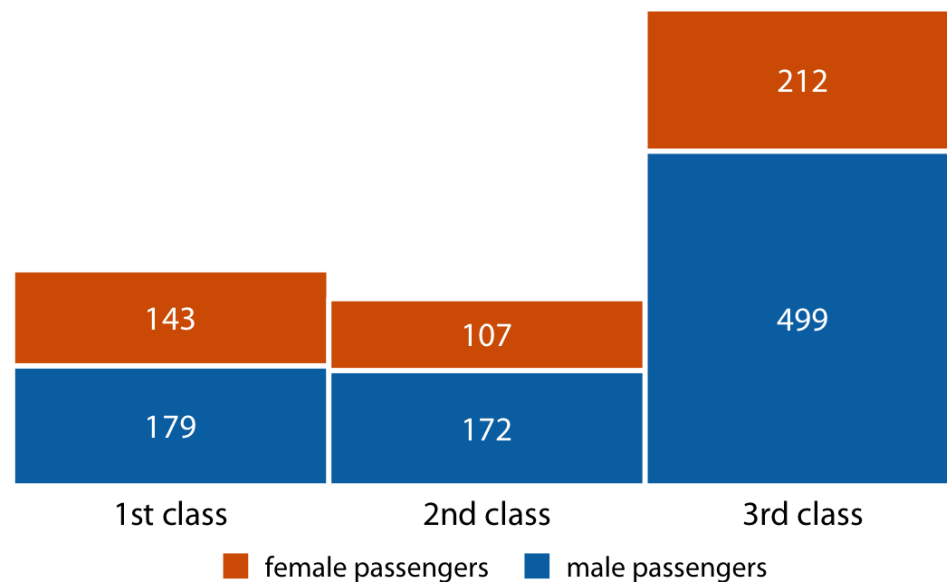
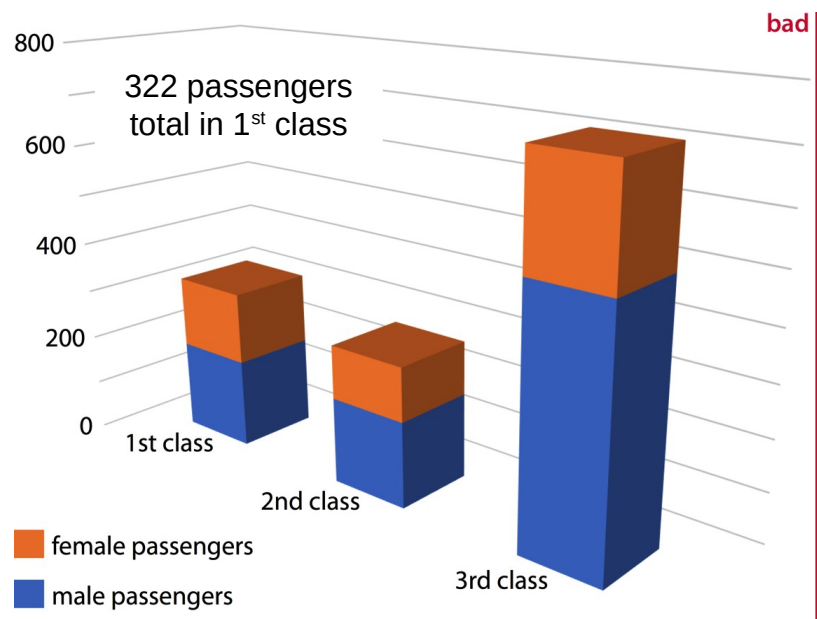
The data-ink ratio (Tufte, 1983)

- Proportion of ink (pixels) that is used to present actual data compared to the total amount of ink used in the entire figure
- For good graphics, maximize the data-ink ratio (within reason)

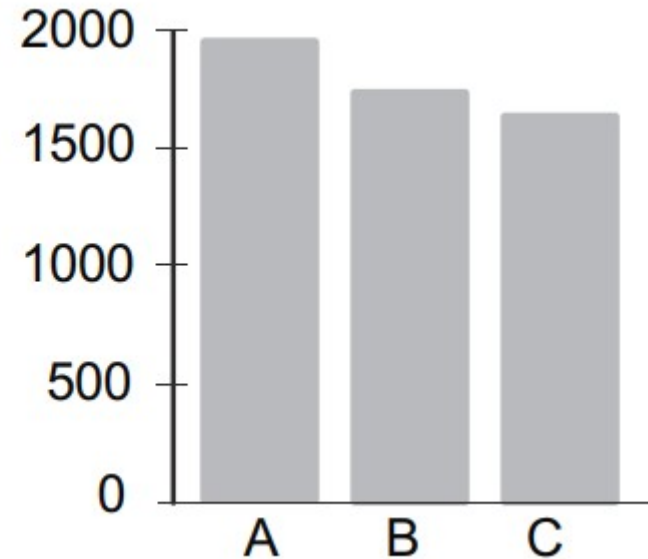
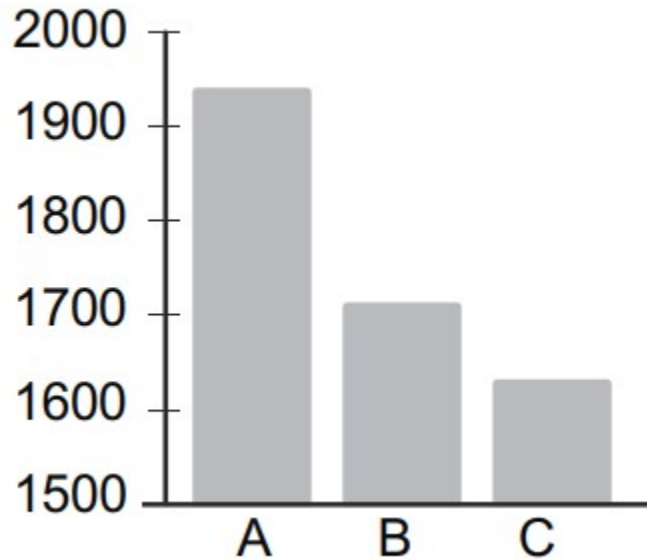


Don't go 3D

3D plots are quite popular, in particular in business presentations but also among academics. They are also almost always inappropriately used. It is rare that I see a 3D plot that couldn't be improved by turning it into a regular 2D figure. In this chapter, I will explain why 3D plots have problems, why they generally are not needed, and in what limited circumstances 3D plots may be appropriate.



Select meaningful axis ranges



Kelleher & Wagener, 2011

- Bars need to start at zero, so that the bar length is proportional to the amount shown
- For some datasets, however, this can be impractical or may obscure key features

Wilke, 2019

Select meaningful axis ranges, and choose an appropriate visualization

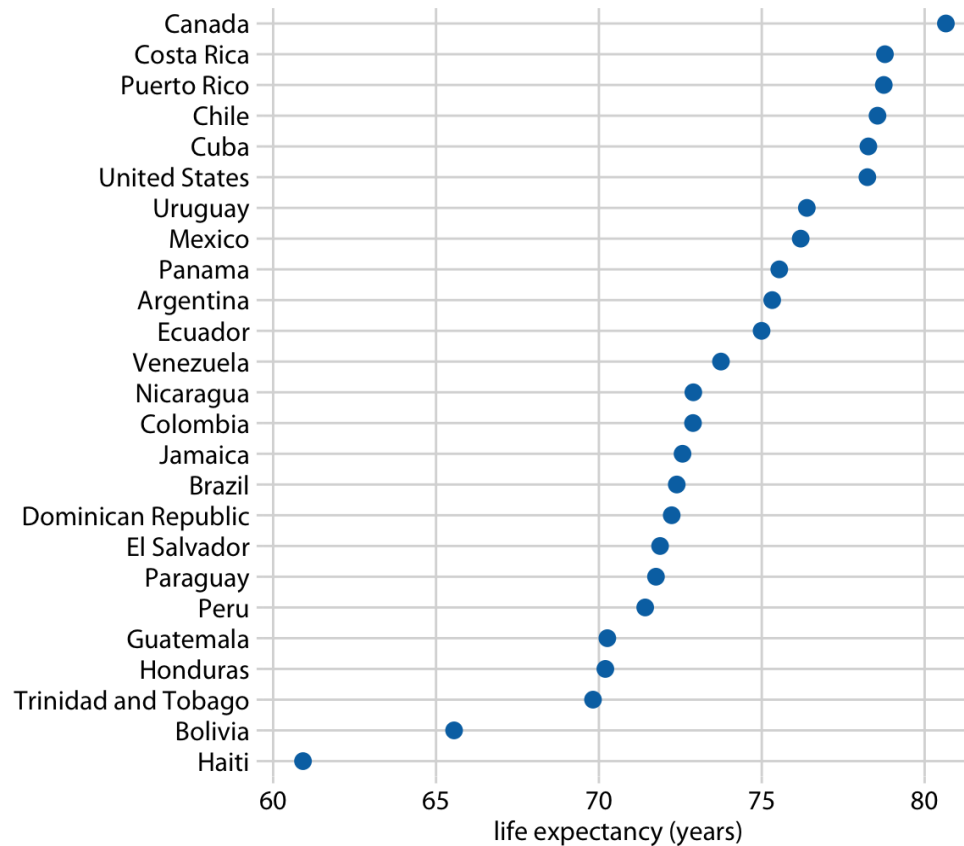
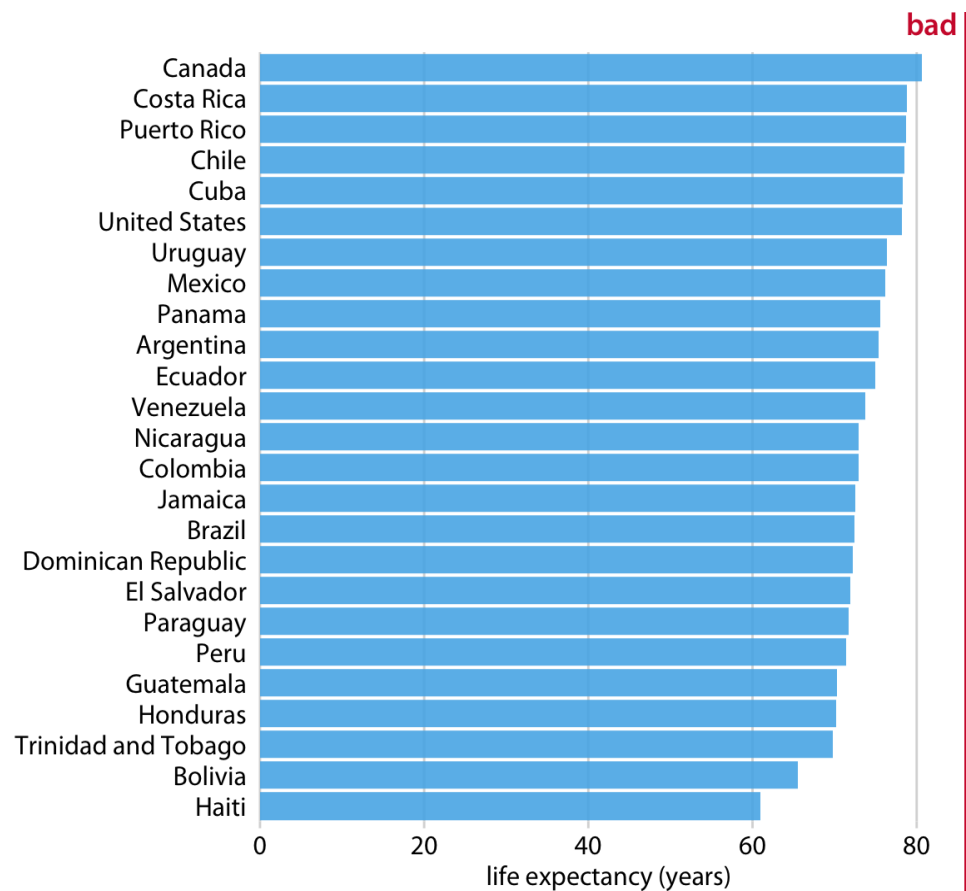
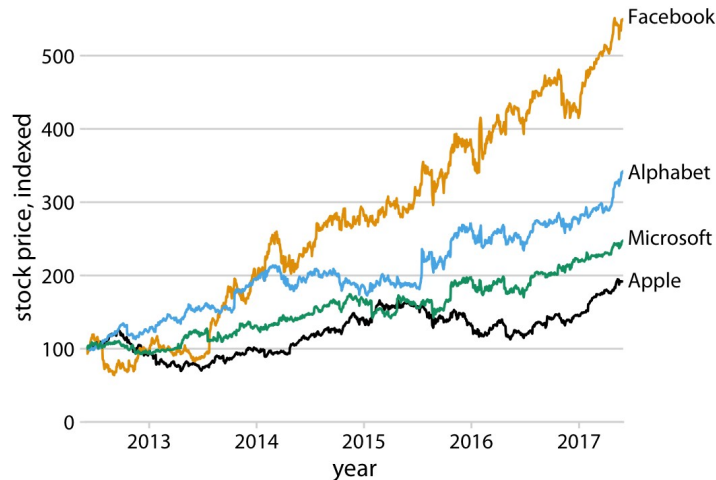
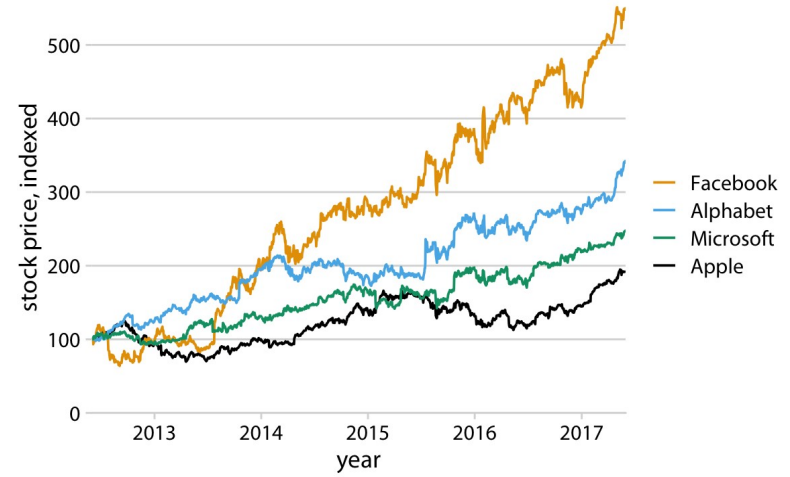
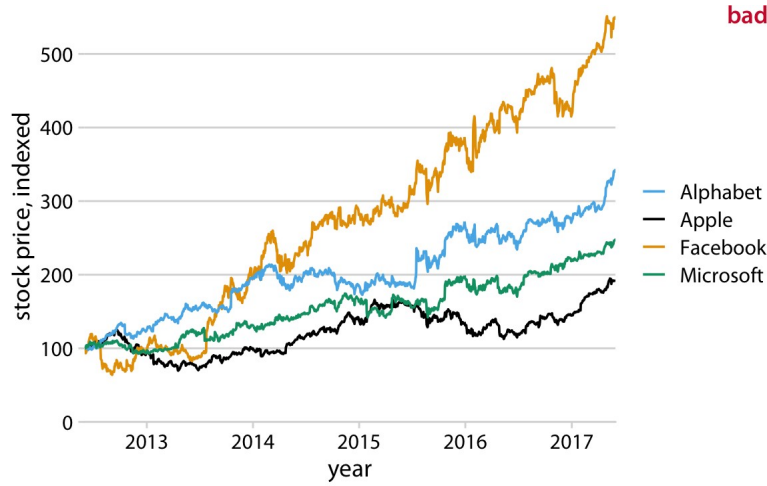
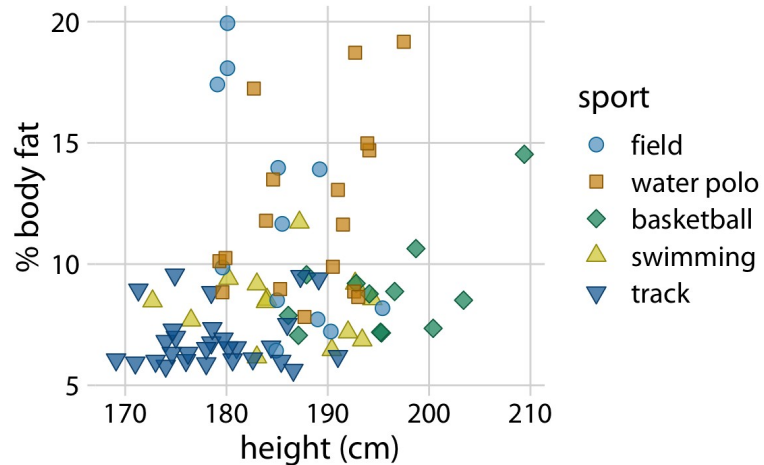
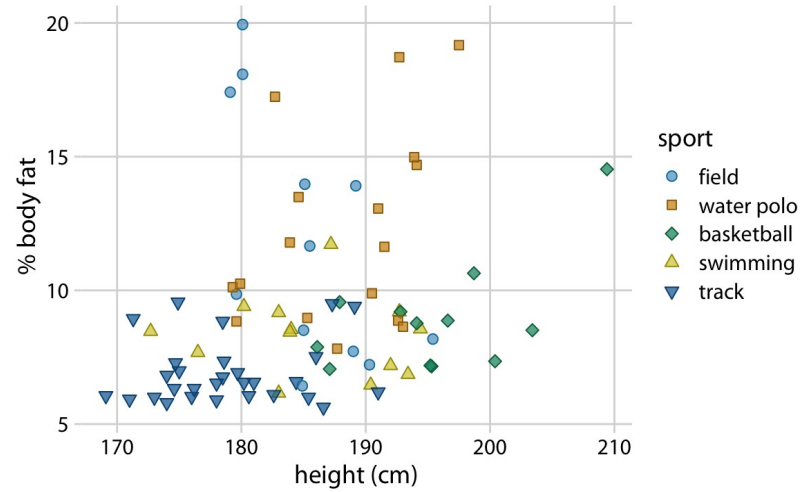
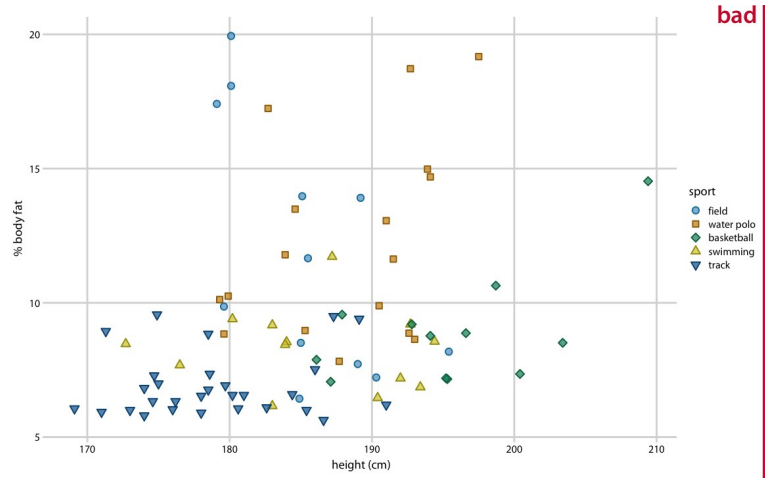


Figure legends



- If there is a clear visual ordering in your data, make sure to match it in the legend
- Whenever possible, design your figures so they don't need a legend

Use larger axis labels



Always look at scaled-down versions of your figures to make sure that the axis labels are appropriately sized.

Part III:

Academic Publishing

Publishing your paper

You have completed your scientific project
and written a nice manuscript about it...

Now what?



Selecting a journal



- Consider Society journals
- Avoid predatory journals!

The publishing process

- Submission
- Peer Review, often necessitating revision and re-submission
- Final editorial decision: Accept or Reject
- Usually quite slow (months or even years)

Preprint servers



Preprints with THE LANCET



- No peer review
- Faster visibility

New publishing models



eLife: eliminating accept/reject decisions after peer review

What is wrong with science publishing?



7:00 / 54:07 • Wichtige Momente >



Reinventing science publishing- Prof. Michael B. Eisen at the University of Alberta

Science communication remains stuck in the 17th century

An Account of the improvement of Optick Glaffes.

The Second, concerns the *Circle of Saturn*, in which he hath observed nothing, but what confirms *Monſieur Chriſtian Hygens de Zolichem* his *Systeme* of that Planet, published by that worthy Gentleman in the year, 1659.

ARTICLE

<https://doi.org/10.1016/j.jmb.2019.05.005>

Structural basis of mitochondrial receptor binding and constriction by DRP1

Kashyap Sahu¹²³, Sun Yi-Ren Wang¹²⁴, Sh Young¹²⁵, Paul V. Thomas¹²⁶, David A. Jones¹²⁷, Janet M. Shaw¹²⁸ & Adam Fox¹²⁹

[illegible][illegible][illegible]

© 1998 Blackwell Science Ltd *Journal of Internal Medicine* 243: 395–401

© 2007 The Authors
Journal compilation © 2007 Blackwell Publishing Ltd



Michael Eiser

Reinventing science publishing- Prof. Michael B. Eisen at the University of Alberta

What is wrong with science publishing?

Restricted

~ .01% of world's population has meaningful access to scientific and medical literature

the vast majority of global researchers, as well as most teachers, health-care workers, journalists and interested members of the public are locked out



What is wrong with science publishing?

VERY Slow

Typical paper takes 9 months from submission to publication.

This delay massively impairs scientific progress, and as major health advance are built on many papers, slows benefits of basic research to public by years.



What is wrong with science publishing?

Profit	Company	Industry
2% of \$173.8 billion		Retail
9% of \$98.7 billion		Automobile
11% of \$110.9 billion		Search
21% of \$229 billion		Computing
37% of \$2.8 billion		Scholarly Publishing
39% of \$608 million	 Taylor & Francis Taylor & Francis Group	Scholarly Publishing

References

- Eisen, M. B. (2021). “Reinventing science publishing- Prof. Michael B. Eisen at the University of Alberta” (Youtube)
- Gustavii, B. (2008). *How to Write and Illustrate a Scientific Paper*. Cambridge, UK: Cambridge University Press.
- Kelleher, C. & Wagener, T. (2011). Ten guidelines for effective data visualization in scientific publications. *Environ. Model. Softw.* 26: 822-827.
- Strunk Jr., W. & White, E. B. *The Elements of Style*. New York City, NY: Macmillan Inc.
- Tufte, E. (1983). *The Visual Display of Quantitative Information*. Cheshire, CT: Graphics Press.
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*. Sebastopol, CA: O'Reilly Media, Inc.